

EXPERIMENT 2: CRYPTARITHMETIC PROBLEM

Aim

To write a Python program to solve the **Cryptarithmic Problem** using Artificial Intelligence techniques.

Objective

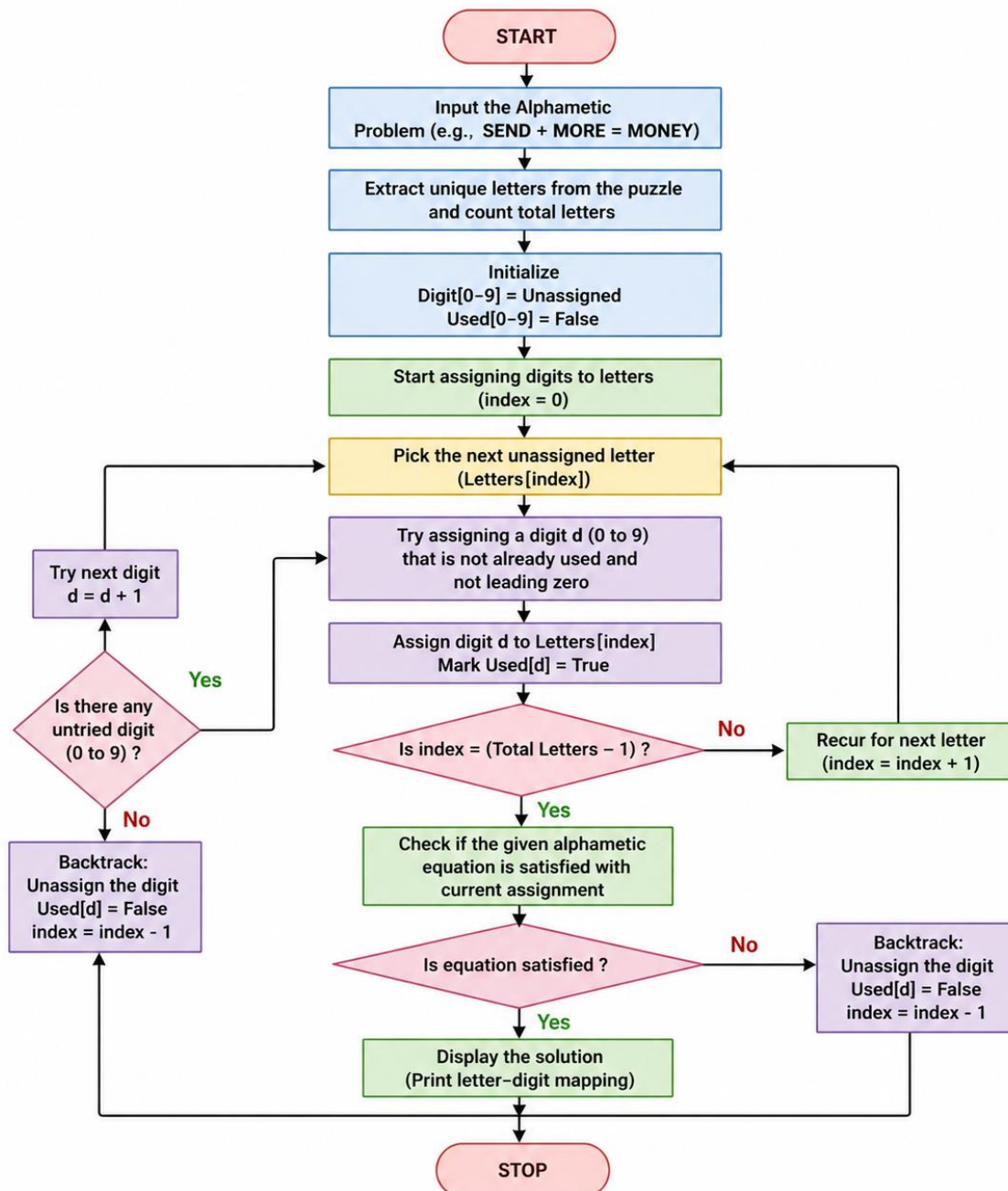
- To understand the concept of constraint satisfaction problems in Artificial Intelligence.
- To assign unique digits to letters and solve arithmetic puzzles.
- To implement a search-based approach for finding a valid solution.

Algorithm

1. Start.
2. Define the letters used in the cryptarithmic puzzle.
3. Assign unique digits (0–9) to each letter.
4. Ensure that no two letters have the same digit.
5. Ensure that the first letter of each word is not assigned zero.
6. Convert the words into numerical values.
7. Check whether the arithmetic equation is satisfied.
8. Display the valid digit assignment.
9. Stop.

Flowchart:

CRYPTARITHMETIC PROBLEM – FLOWCHART



Code (Python)

```
from itertools import permutations
```

```
def cryptarithmic():
```

```
    letters = ('S', 'E', 'N', 'D', 'M', 'O', 'R', 'Y')
```

```
for values in permutations(range(10), len(letters)):
```

```
    mapping = dict(zip(letters, values))
```

```
    # Leading letters should not be zero
```

```
    if mapping['S'] == 0 or mapping['M'] == 0:
```

```
        continue
```

```
    SEND = (1000 * mapping['S'] +
```

```
            100 * mapping['E'] +
```

```
            10 * mapping['N'] +
```

```
            mapping['D'])
```

```
    MORE = (1000 * mapping['M'] +
```

```
            100 * mapping['O'] +
```

```
            10 * mapping['R'] +
```

```
            mapping['E'])
```

```
    MONEY = (10000 * mapping['M'] +
```

```
             1000 * mapping['O'] +
```

```
             100 * mapping['N'] +
```

```
             10 * mapping['E'] +
```

```
             mapping['Y'])
```

```
    if SEND + MORE == MONEY:
```

```
        print("Solution Found:")
```

```
        print("SEND =", SEND)
```

```
        print("MORE =", MORE)
```

```
        print("MONEY =", MONEY)
```

```
        print("Mapping =", mapping)
```

```
        break
```

cryptarithmic()

Output



```
IDLE Shell 3.12.4
File Edit Shell Debug Options Window Help
Python 3.12.4 (tags/v3.12.4:8e8a4ba, Jun 6 2024, 19:30:16) [MSC v.1940 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:/Users/sandh/OneDrive/Desktop/experiment 1.py
Solution Path:
(0, 0)
(0, 3)
(3, 0)
(3, 3)
(4, 2)
>>>
===== RESTART: C:/Users/sandh/OneDrive/Desktop/experiment 2.py =====
Solution Found:
SEND = 9567
MORE = 1085
MONEY = 10652

Letter Mapping:
S = 9 E = 5 N = 6 D = 7
M = 1 O = 0 R = 8 Y = 2
>>>
Ln: 21 Col: 0
```

Result

Thus, the Python program for solving the **Cryptarithmic Problem (SEND + MORE = MONEY)** was executed successfully, and the correct digit assignment was obtained.

Conclusion

The experiment helped in understanding constraint satisfaction problems in Artificial Intelligence. The cryptarithmic puzzle was solved successfully by applying a search technique to find valid letter-to-digit mappings.